Dual-Stream Class-Adaptive Network for Semi-Supervised Hyperspectral Image Classification

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***Abstract*— Semi-supervised classification of remote sensing (RS) hyperspectral image (HSI) aims at exploiting both labeled and unlabeled samples for accurate land cover recognition. However, imbalanced data distribution and different classification difficulties negatively affect classification performance. Focused on this, a novel dual-stream class-adaptive network (DSCA-Net) is proposed for semi-supervised HSI classification, in this article. First, a superpixel-guided label propagation (SGLP) module is introduced to alleviate the negative effect of imbalanced data distribution. Specifically, approximate estimation of labels for unlabeled samples is achieved via superpixel-wise similarity measure and label propagation, so that equal sampling is applied to each class. Then, a consistency regularization-based dual-stream network is constructed, which shares the same encoder for feature representation of either labeled or unlabeled samples. Based on this, two distinct classifiers are designed to force similar predictions that can be achieved for various perturbed versions of the same unlabeled sample, thereby allowing unlabeled samples to train the model in a supervised manner. Finally, since different classes always have various degrees of learning difficulty, equal treatment may lead to overfitting of "easy" classes and biased prediction of "hard" classes. Unlike the traditional selection of unlabeled samples with a fixed threshold, dynamic class-adaptive thresholds are calculated according to the learning status of the model. In this manner, a higher threshold is assigned to "easy" classes to reduce sample redundancy, and a lower threshold is set for "hard" classes to select more samples. Experiment results demonstrate the effectiveness and superiority of the proposed method. Codes are available at https://github.com/luting-hnu/DSCA-Net.**

***Index Terms*— Consistency regularization, deep learning (DL), hyperspectral image (HSI) classification, semi-supervised learning (SSL), superpixel segmentation.**

I. Introduction

HYPERSPECTRAL sensors with increasing spectral resolution have been able to accurately identify land covers, with the rapid development of modern remote sensing (RS) technology. Thus, hyperspectral image (HSI) classification has attracted much attention, which plays important roles in precision agriculture [1], mineralogy, military defense [2], urban development [3], and other fields [4]. However, due to the high-dimensional spectral bands and complex spatial structure of HSI, how to learn discriminative feature representation is still a challenging task, especially when limited training samples are available.

In recent years, deep learning (DL) methods relying on neural networks have provided a promising way for RS image classification. Among them, the convolutional neural network (CNN) model is one research hotspot, due to its powerful ability to extract spatial features by convolutional operations [5], [6], [7], [8], [9]. Chen et al. [10] first used CNN to extract regularized deep features in HSI classification. Considering HSIs always have different scales of structures, Zhang et al. [11] constructed a multiscale dense network to extract fine and coarse features in the horizontal and vertical directions, respectively. Zhang et al. [12] combined CNN with morphological transformation and spatial-logical aggregation techniques for tree species classification using hyperspectral imagery. More lately, many scholars attempted to introduce visual attention mechanisms to CNN, to push the networks to focus on more valuable information for representation learning of higher discriminative features. As variations of spectral–spatial attention, Liu et al. [13] elaborated a scaled dot-product central attention and multiarea target attention [14] to enhance HSI classification behavior. Lu et al. [15] introduced a new band-grouping guided multiattention network to learn group-specific spectral–spatial features. In addition, some creative works based on multi-modal or cross-scene classification can also combine image information from diverse modalities or sources to tackle the inconsistency of feature representations and the insufficient expression of semantic relationships [16], [17]. Furthermore, a number of advanced network structures or new DL models in the computer vision domain were proposed to improve HSI classification performance. For example, transformers have been introduced into the HSI classification task owing to their good ability to characterize long-term relationships [18], [19]. Generative adversarial network (GAN) have also attracted much attention due to the unique mutual adversarial training mode [20], [21].

However, the above DL methods belong to supervised learning paradigm, in which performance is negatively affected

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when there are insufficient training samples [22]. To eliminate the dependence on labeled samples, some DL methods began to explore semi-supervised learning (SSL) strategy in HSI classification task [23], [24], [25], [26]. Specifically, SSL methods seek to improve performance by using both the labeled and unlabeled samples [27], which can be broadly classified into three categories, including generative-based models, graph-based methods, and self-training methods. Among them, generative-based models aim to obtain contextual information about unlabeled samples via autoencoder (AE) model. For example, Zhou et al. [28] used the spectral and spatial features of HSI to pre-train two stacked AEs separately, then enlarged the initial training set in an SSL co-training fashion. Jia et al. [29] integrated an AE module and a Siamese network to investigate the information in a large amount of unlabeled data, respectively, then fine-tuned it with a limited labeled sample set. As to the graph-based methods, they achieve the classification objective through capturing the intrinsic structure revealed by the labeled and unlabeled samples together. Kotzagiannidis and Schönlieb [30] proposed a multistage edge-efficient SSL graph network for HSI classification, which utilized labeled samples by embedding pseudo-labeled features in graph construction. Xi et al. [31] proposed a new cross-scale graph prototype layer to enhance the discrimination and representativeness of node features and prototypes for semi-supervised HSI classification. For the self-training format, it requires iteratively updating high-confidence unlabeled samples (i.e., pseudo-labels) and combines with a few labeled samples to train a classifier [32]. Feng et al. [33] proposed a semi-supervised long-tail learning method that determined the labels of unlabeled samples based on spatial neighborhood information and re-determined them with new criteria to improve the accuracy of pseudo-labeling. However, the generative-based models rely on appropriate constraints to produce highly confidence pseudo-labeled samples, which is difficult to satisfy in practice. The graph-based methods can well construct the connection between labeled and unlabeled samples for label propagation, which is always computationally intensive. In contrast, the self-training format has been widely studied due to its simple principle and promising performance.

Based on the above analysis, we attempted to design a new semi-supervised DL method based on the CNN model and the self-training format, in this article. There are two main issues we would like to alleviate here, i.e., the imbalanced data distribution and different learning difficulties of various categories. In the self-training format, a portion of the data needs to be randomly selected as an unlabeled sample set (ULS). However, due to the fact that data distribution is always imbalanced [34], the category with a small amount may not be selected, leading to a biased estimation of classification. In addition, a high threshold is required for the self-training format to select high-confidence pseudo-labeled samples, which is usually set to be a high value, e.g., 0.95 [35]. In fact, different categories should have different learning difficulties [36]. As shown in Fig. 1(a), setting the same threshold for all pseudo-labeled samples results in the fact that only a few samples or even no sample can exceed such a high threshold for the classes with high learning difficulties. In contrast, a dynamic class-adaptive threshold is necessary for different classes, as shown in Fig. 1(b).

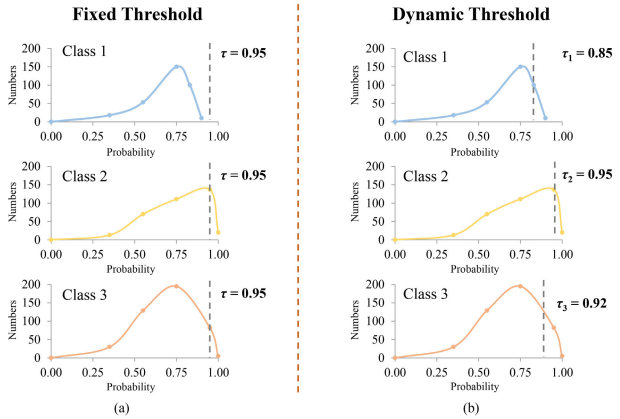


Fig. 1. Illustration of the comparison between fixed threshold and dynamic thresholding strategies. It can be seen that for class 1, none of the samples can exceed the threshold of 0.95.

Focused on the negative effects of imbalanced data distribution and various learning difficulties, we proposed a novel dual-stream class-adaptive network (DSCA-Net) for semi-supervised HSI classification. First, to alleviate the imbalanced data distribution, a superpixel-guided label propagation (SGLP) module is proposed to propagate known labels via superpixel-wise similarity measure and label propagation, providing a category-balanced unlabeled training set for subsequent training. Previously, superpixel-based techniques were useful in HSI classification. Zhao et al. [37] proposed a superpixel-guided deformable convolution to adaptively extract features based on the spatial structure of HSI. Jia et al. [38] proposed a superpixel-level weighted label propagation method without any parameter tuning. They aimed to achieve continuous iterative feature learning or label updating at the superpixel level, which costs extensive computation. Unlike these works, the module we proposed performs cosine similarity measure and label propagation just once before network training, costing a low amount of computation. Then, depending on consistency regularization, two distinct classifiers are designed to force similar predictions to be achieved for various perturbed versions of the same unlabeled sample, thereby allowing unlabeled samples to train the model in a supervised manner. Most importantly, for the insufficiency of the traditional fixed threshold method that sets the same threshold for each class [39], a class-adaptive dynamic thresholding (CADT) strategy is proposed. Here, the threshold corresponding to each class is obtained from a difficulty estimation computed based on the learning status of the model. Furthermore, a nonlinear mapping function is designed to project the difficulty estimation to the ultimate adaptive threshold of each class. In summary, the main contributions of our method are as follows.

1) A novel dual-stream semi-supervised DL network is proposed for HSI classification. In specific, two streams share the same encoder for similar feature representation and adopt two classifiers for distinct label estimation. By optimizing both supervised and unsupervised

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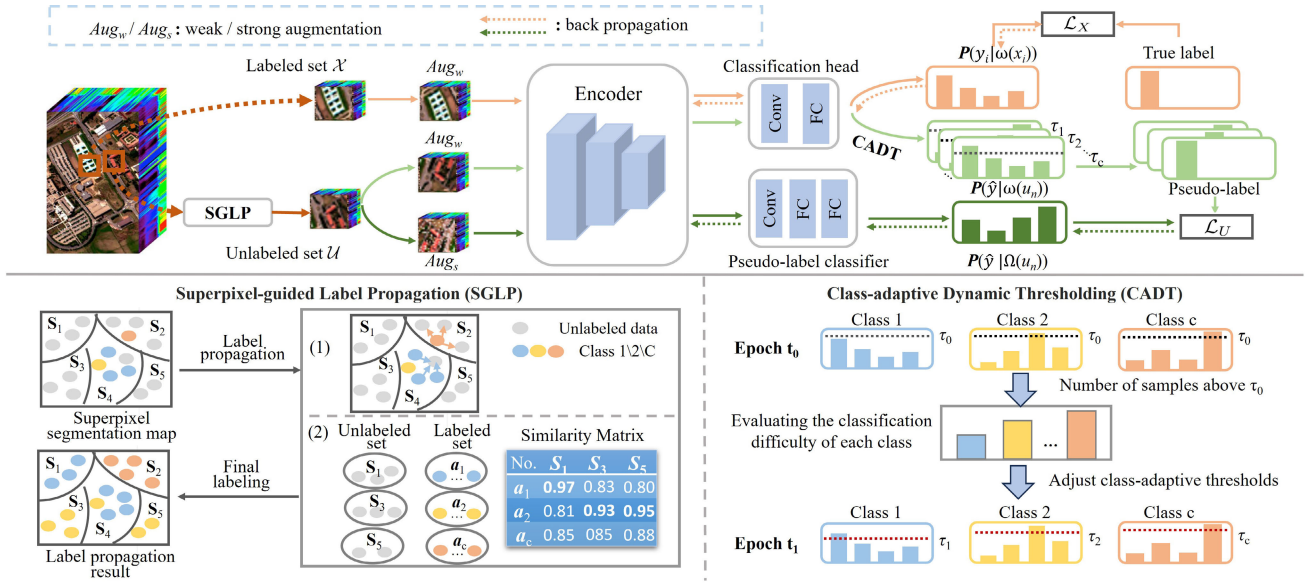


Fig. 2. Flowchart of the proposed method DSCA-Net.

cross-entropy loss jointly, the network model is trained to learn more discriminative features from both labeled and unlabeled samples.

2) To mitigate the negative effect of the imbalanced data distribution, an SGLP module is designed to rapidly generate the approximate estimation of possible labels for all of the unlabeled pixels. Based on the similarity measure, the limited known labels are propagated superpixel-by-superpixel. Depending on this, we can select an equal number of unlabeled samples from each class to construct a balanced ULS.
3) To adaptively determine thresholds for better selection of highly confident pseudo-labeled samples, a CADT strategy is proposed here. Considering different classes have different learning difficulties, we estimate their classification difficulties and use the estimated values to adjust the thresholds. This can help to alleviate the overfitting of "easy" classes and the biased prediction of "hard" classes.

The remainder of this article is organized as follows. Section II presents the details of the proposed method. Section III gives experimental results and analysis of several publicly available datasets. Finally, we summarize our work in Section IV.

II. Methodology

First of all, the DSCA-Net carefully picks up unlabeled samples with the developed SGLP module, to alleviate the negative impact of imbalanced data distribution. Then, we construct a dual-stream structure with the consistency regularization concept [40] that similar predictions are achieved for various perturbed versions of the same unlabeled sample. In particular, considering the diverse learning difficulties of different classes, we design the CADT strategy to adaptively tone thresholds for reliable sample selection. That is, a higher threshold is assigned to "easy" class, and vice versa. The final classification network is optimized through the iterative learning of unlabeled samples using a hybrid loss function. The flowchart of the proposed method DSCA-Net is shown in Fig. 2. Detailed descriptions are given below.

A. Superpixel-Guided Label Propagation

Reasonable selection of unlabeled samples is important for SSL since random selection leads to imbalanced sampling due to the imbalanced data distribution. To alleviate the negative impact of the imbalanced distribution, an SGLP module is introduced in this section. That is, we propagate the limited known labeled samples to the whole HSI under the guidance of spectral–spatial similarity-based superpixel segmentation. With those propagated labels, equal sampling for different classes is achieved.

Here, we use simple linear iterative clustering (SLIC) [41] to segment the HSI into a set of superpixels which are non-overlapping homogeneous regions with different shapes and sizes. Then, these superpixels can be divided into two groups, according to whether there are labeled samples within them. As shown in Fig. 2, one group consists of superpixels including labeled samples, such as $\boldsymbol{S}_2$ and $\boldsymbol{S}_4$. Superpixels have the characteristic that pixels within one superpixel are highly similar and probably belong to the same class. Therefore, we count the number of samples with known labels and propagate the labels corresponding to the largest number to other unlabeled samples in the same superpixel. The other group only contains superpixels without known labels, such as $\boldsymbol{S}_1$, $\boldsymbol{S}_3$, and $\boldsymbol{S}_5$ in Fig. 2. We assign labels to these superpixels with a superpixel-level nearest neighbor algorithm. Specifically, distances between each superpixel and each class of labeled samples are calculated from their average spectrums that are denoted by $\boldsymbol{s}_i$ and $\boldsymbol{a}_j$, respectively. The formula is expressed as

$$D_{ij} = \cos\theta = \frac{\boldsymbol{s}_i \cdot \boldsymbol{a}_j}{\|\boldsymbol{s}_i\| \|\boldsymbol{a}_j\|} \tag{1}$$

where $\|\cdot\|$ denotes the modulus of the average spectrums. For the superpixel without known labels $\{\boldsymbol{S}_i\}_{i=1,3,5}$, we can obtain

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distances $\{D_{ij}\}_{j=1,2,\ldots,C}$, where C is the number of classes. The larger D_{ij} means $\boldsymbol{s}_i$ and $\boldsymbol{a}_j$ are more similar, and vice versa. Therefore, the label c corresponding to the maximum distance D_{ij} is assigned to superpixel $\boldsymbol{S}_i$ and then propagated to all pixels in it.

Once all pixels in the HSI own propagated labels under the guidance of superpixels, an approximate category-balanced set can be achieved by equally selecting unlabeled samples from each class. After that, the labeled set $\mathcal{X}$ and the unlabeled set $\mathcal{U}$ are fed into the dual-stream structure with consistency regularization for further training.

B. Consistency Regularization-Based Dual-Stream Structure

Consistency regularization [42] provides a flexible training strategy for unlabeled samples, based on the idea that similar predictions are achieved for various perturbed versions of the same unlabeled sample. To make full use of unlabeled samples for training, we construct a dual-stream structure with consistency regularization. One is the supervised stream while the other one is the unsupervised stream. The two streams, respectively, use weak augmentation $\omega(\cdot)$ and strong augmentation $\Omega(\cdot)$ to achieve different perturbed versions of unlabeled samples. Their prediction results are then enforced to be similar, which can use the pseudo-labels of the supervised stream to help the training of the unsupervised stream. Detailed descriptions are shown below.

The supervised stream contains one weak augmentation operation, an encoder, and a classification head. The encoder is composed of three 2-D convolutional layers with 3×3 kernel size, the number of channels is 64, 128, 256, as well as the batch normalization layer and the ReLU activation function layer. The classification head consists of a 2-D convolutional layer with 1×1 kernel size and one fully connected layer. First, the stream is trained by optimizing a supervised cross-entropy loss $\mathcal{L}_X$ conducted on the labeled set $\mathcal{X}$. Then, the unlabeled sample $u_n \in \mathcal{U}$ after weak augmentation is fed into the encoder and the classification head to predict estimation probability $\boldsymbol{P}(\hat{y}|\omega(u_n))$. Here, $\hat{y}$ is an artificial label (i.e., pseudo-label). High probability indicates that the pseudo-label is reliable for the unlabeled sample [43], and vice versa. Based on this, we assign pseudo-labels to the unlabeled samples with high prediction probabilities. As a result, a thresholding operation is used to generate pseudo-labels by

$$\hat{y} = \arg\max \boldsymbol{P}\left(\hat{y}|\omega(u_n)\right) \quad \text{s.t.,} \max \boldsymbol{P}\left(\hat{y}|\omega(u_n)\right) > \tau \tag{2}$$

where τ is a predefined threshold. Obviously, the higher the τ, the less the samples with pseudo-labels, and vice versa. Therefore, τ is crucial for the generation of pseudo-labels. In particular, we design a CADT strategy to automatically determine the thresholds for different categories, which will be further introduced in Section II-C.

The unsupervised stream is specially designed for unlabeled samples with pseudo-labels, which consists of a strong augmentation operation, an encoder, and a classifier. The pseudo-label classifier (PLC) is composed of a 2-D convolutional layer with 1×1 kernel size and two fully connected layers. Moreover, the encoders from two streams share the same parameters. After the input of unlabeled samples, the classifier outputs the classification probability $\boldsymbol{P}(\hat{y}|\Omega(u_n))$, where $\Omega(u_n)$ is the strong augmentation of u_n. The training of this stream is difficult since no supervised information is available. Based on the idea of consistency regularization, the pseudo-labels can be used as reliable supervised information to guide the training of this stream. To this end, a cross-entropy loss $\mathcal{L}_U$ is constructed. Finally, the alternating training of two streams leads to the incremental optimization of the shared encoder and classification head. As a result, the implicit classification information of the unlabeled set $\mathcal{U}$ can be fully exploited to improve classification performance.

C. Class-Adaptive Dynamic Thresholding

As shown in the last section, threshold τ is crucial for the exploitation of unlabeled samples, especially in the field of SSI. In practice, some classes are easier to classify and have high classification probabilities, and vice versa [44]. Given a fixed threshold τ, the network will pay attention to more samples from "easy" classes while ignoring samples from "hard" classes. As a result, the training of the network may suffer from overfitting for "easy" classes and bias prediction for "hard" classes, as shown in Fig. 1.

To settle this issue, we design a new CADT strategy to estimate various thresholds $\{\tau_c\}_{c=1,2,\ldots,C}$ for different classes. That is, "easy" classes with numerous samples are assigned to larger thresholds, and vice versa. To this end, we first evaluate the classification difficulty σ_c of the cth class using the number of samples whose classification probabilities are larger than the class-adaptive threshold $\tau \in \{\tau_c\}_{c=1,2,\ldots,C}$ [45]. Given a very high initial threshold τ_0, the σ_c is obtained by

$$\sigma_c = \sum_{n=1}^{N_{\mathcal{U}}} \left[\max \boldsymbol{P}\left(\hat{y}|\omega(u_n)\right) > \tau_0\right] \quad \text{s.t.,} \arg\max_{\hat{y}} \boldsymbol{P}\left(\hat{y}|\omega(u_n)\right) = c \tag{3}$$

where $[\cdot]$ is the Iverson bracket with the value of 1 if the condition in square brackets is satisfied, and 0 otherwise. $N_{\mathcal{U}}$ is the total number of unlabeled samples. Then, we normalize σ_c to range $[0, 1]$ by $\beta_c = (\sigma_c / \max_c \sigma_c)$. It can be seen that classes with fewer samples correspond to lower β_c, and vice versa. To ensure that each class can obtain sufficient learnable samples, we should increase the sample number for classes with a few samples and reduce the information redundancy for classes with numerous samples. In this way, we could alleviate the overfitting of classes with numerous samples and biased prediction of the classes with a few samples, keeping the number of each class in a balanced state and facilitating the discriminability and robustness of the model.

Next, we construct the weight with the mapping $\varphi(\beta_c)$, to scale the initial threshold τ_0. As a result, the class-adaptive threshold τ_c can be achieved by

$$\tau_c = \varphi(\beta_c) \cdot \tau_0. \tag{4}$$

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According to (4), threshold τ_c is positively associated with the weights $\varphi(\beta_c)$. Due to that "easy" classes with numerous samples have large thresholds and "hard" classes with a few samples have small thresholds, the mapping $\varphi(\cdot)$ should be a monotone increasing function for $\beta_c \in [0, 1]$. Besides, for "hard" classes, the weights $\varphi(\beta_c)$ decrease more and more fast as β_c decreases, to assign more pseudo-labels for learning. For "easy" classes, the weights $\varphi(\beta_c)$ are all very close to the maximum value 1, which means that the weights increase slowly as β_c rises. Therefore, the mapping $\varphi(\cdot)$ should be a nonlinear convex function for $\beta_c \in [0, 1]$. Based on the above considerations, we can use a simple parabolic function with the maximum value of 1 (obtained when variable $\beta_c = 1$) for mapping. The formula can be

$$\varphi(\beta_c) = -\lambda(\beta_c - 1)^2 + 1 \tag{5}$$

where $\lambda \in (0, 1]$ is used to determine the value for the case of $\beta_c = 0$, and is discussed in the ablation experiments.

Through the above steps, we achieve the purpose of setting different thresholds for different classes while obtaining high-quality pseudo-labels. As a result, the generalization ability and the robustness of the model are improved, as well as the final classification performance.

D. Overall Loss

The loss function of DSCA-Net consists of two parts: the supervised cross-entropy loss on labeled samples $\mathcal{L}_X$, and the unsupervised cross-entropy loss on unlabeled samples $\mathcal{L}_U$. The mathematical formula is expressed as follows:

$$\mathcal{L} = \mathcal{L}_X + \mathcal{L}_U \tag{6}$$

where the $\mathcal{L}_X$ and $\mathcal{L}_U$ are expressed, respectively, as

$$\mathcal{L}_X = \frac{1}{N_{\mathcal{X}}} \sum_{i=1}^{N_{\mathcal{X}}} H(y_i, \boldsymbol{P}(y_i|\omega(x_i))) \tag{7}$$

$$\mathcal{L}_U = \frac{1}{N_{\mathcal{U}}} \sum_{n=1}^{N_{\mathcal{U}}} H\left(\hat{y}, \boldsymbol{P}\left(\hat{y}|\Omega(u_n)\right)\right) \quad \text{s.t., } \max \boldsymbol{P}\left(\hat{y}|\omega(u_n)\right) > \tau_{\hat{y}} \tag{8}$$

where $H(\cdot)$ is the cross-entropy function. $N_{\mathcal{X}}$ and $N_{\mathcal{U}}$ are the total number of labeled samples and unlabeled samples, respectively.

III. Experimental Results and Discussions

In this section, we describe the conducted experiments to demonstrate the effectiveness of the proposed DSCA-Net for HSI classification. First, we briefly describe the datasets used in the experiments, followed by the experimental setup and the ablation experiments. Then, we perform comparison experiments of our method with some state-of-the-art methods.

A. Dataset Description

Four widely used HSI datasets are utilized in the experiments, including the University of Pavia (PaviaU), Houston 2013, Kennedy Space Center (KSC), and Botswana datasets. We will give a short introduction to the datasets below.

TABLE I
Sample Distributions on the PaviaU and Houston 2013 Datasets

PaviaU				Houston 2013			
No.	Class Name	Train	Test	No.	Class Name	Train	Test
1	Asphalt	10	6621	1	Healthy grass	10	1241
2	Meadows	10	18639	2	Stressed grass	10	1244
3	Gravel	10	2089	3	Synthetic grass	10	687
4	Trees	10	3054	4	Trees	10	1234
5	Metal Sheets	10	1335	5	Soil	10	1232
6	Bare Soil	10	5019	6	Water	10	315
7	Bitumen	10	1320	7	Residential	10	1258
8	Bricks	10	3672	8	Commercial	10	1234
9	Shadows	10	937	9	Road	10	1242
	Total	90	42686	10	Highway	10	1217
				11	Railway	10	1225
				12	Parking Lot 1	10	1223
				13	Parking Lot 2	10	459
				14	Tennis Court	10	418
				15	Running Track	10	650
					Total	150	14879

1) PaviaU: This dataset was acquired by the ROSIS-03 sensor of the University of Pavia, Italy. It consists of 103 spectral bands in the wavelength range 0.43 to 0.86 μm, with nine land cover classes, and its spatial size is 610 $\times$ 340 pixels. A total of 42 776 labeled samples are available, without including the background classes.

2) Houston 2013: This dataset was collected by the ITRES CASI-1500 sensor in and around the University of Houston. It contains 144 spectral bands in the wavelength range of 0.364 to 1.046 μm and 15 land cover classes with a spatial size of 349 $\times$ 1905 and a spatial resolution of 2.5 m/pixel. A total of 15 029 labeled samples are available, without the background categories.

3) KSC: This dataset was taken by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor at Kennedy Space Center, Florida. It contains 176 bands with water vapor noise bands removed, and the spatial resolution is 18 m with 512 $\times$ 614 spatial sizes. There are 13 categories and 5211 labeled pixels are available.

4) Botswana: This dataset was acquired by the Hyperion sensor on the EO-1 satellite over the Okavango Delta, Botswana. Once water vapor noise bands have been removed, it contains 145 spectral bands with 30 m spatial resolution and 1476 $\times$ 256 spatial sizes. A total of 2989 labeled samples and 14 categories are available, without the background categories.

B. Experiment Setup

In our experiments, the proposed DSCA-Net method is implemented on a single NVIDIA GeForce RTX 3090 GPU with the PyTorch platform. We randomly selected ten labeled samples for each class as the training set and the rest as the test set. All the experiments are conducted by ten times and the average accuracies are recorded. The objective metrics include overall accuracy (OA), average accuracy (AA), class accuracy (CA), and Kappa coefficient (κ).

The details of the training and test samples of the PaviaU and Houston 2013 datasets are listed in the second column of Table I. The spatial size of the input HSI cube is 24 $\times$ 24, the

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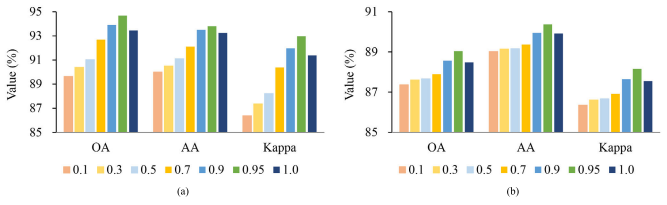


Fig. 3. Influence of the parameter τ_0 on the classification performance of the proposed method in terms of (a) PaviaU and (b) Houston 2013.

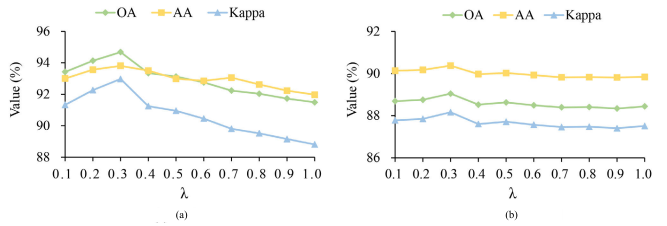


Fig. 4. Influence of the parameter λ on the classification performance of the proposed method in terms of (a) PaviaU and (b) Houston 2013.

optimizer is Adam, the learning rate is 1e-3, and the number of epochs is 200. Besides, the data values are normalized to the range along the spectral dimension [0, 1]. The CADT strategy determines the assignment of the pseudo-labels, and the hyperparameters τ_0 and λ directly affect the class-adaptive thresholds. So, we set $\tau_0 = 0.95$ and $\lambda = 0.3$, which will be detailed in the parameter discussions. Most classification methods use data augmentation to improve the robustness of the model and prevent overfitting. Following them, we utilize data augmentation in the DSCA-Net as well as in the DL-based comparison method. Specifically, weak augmentation $\omega(\cdot)$ indicates random vertical and horizontal flip, strong augmentation $\Omega(\cdot)$ adds random noise augmentation based on weak augmentation. The number of superpixel segmentation M is estimated by

$$M = \frac{H \cdot W}{\varepsilon} \tag{9}$$

where the ε affects the number of pixels in each superpixel. Empirically, ε is set to 200 for the Houston 2013 dataset, and 50 for the other datasets. H and W are the length and the width of the HSI, respectively.

C. Parameter Discussion

For the proposed DSCA-Net method, there are two hyperparameters which will be discussed here, including the threshold upper limit τ_0 and the parameter λ in the nonlinear mapping function. The two parameters affect the quantity and quality of pseudo-labels, thus affecting the classification ability of the model.

1) Threshold Upper Limit τ_0: Experiments are conducted using a grid search to find the most suitable size of τ_0 values, $\tau_0 \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95, 1.0\}$. Fig. 3 illustrates the effect of τ_0 on the classification performance. The parameter τ_0 determines the upper threshold value for selecting highly confident pseudo-labeled samples. In general, as τ_0 increases, the values of the three objective metrics (OA, AA, and κ) become higher. It should also be noted that there are small decrease in OA, AA, and κ when $\tau_0 = 1.0$. When the upper value is too low, the quantity of pseudo-labels capable of exceeding this value becomes larger, meanwhile, the confidence level deteriorates sharply. When the upper is fixed at a high value, the confidence of pseudo-labels is guaranteed, but the number of pseudo-labels is drastically decreased. Based on the above analysis, τ_0 should be neither too large nor too small. In this work, τ_0 is set to 0.95 to achieve good results.

2) Parameter λ in the Nonlinear Mapping Function: It is known that the prediction ability of the model will be gradually improved along with the training process, which always has a comparatively weak capability in the initial training period. Based on this consideration, it is desirable to set a lower threshold to mitigate the negative impact of reckless prediction by the model initially. This can be achieved by adjusting the parameter λ. Fig. 4 shows the effect of different λ on the classification results. For the PaviaU and Houston datasets, the OA, AA, and κ values all reach the highest values when $\lambda = 0.3$. If the value of λ is too small, the threshold is always at a high value, contrary to our proposed class-adaptive thresholding strategy. Conversely, the lower limit of the threshold is too small thus leading to a decrease in the quality of the pseudo-labels. To achieve satisfied classification results, we set $\lambda = 0.3$ in all the following experiments.

D. Ablation Studies

The ULS, PLC, SGLP module, and CADT strategy are essential components in our method. We design ablation experiments to verify their effectiveness. The experimental results on two different datasets are shown in Table II, and the best results in the table have been indicated in bold.

ULS: The ULS is removed to investigate the effect of the added unlabeled samples on the proposed network. The method is named w/o (i.e., without) ULS. The results are shown in Table II. As can be seen, DSCA-Net achieves better values on three comprehensive evaluation metrics of OA, AA, and κ. Specifically, the OA of DSCA-Net is 3.78% and 1.31% higher than the OA w/o ULS on the two datasets, verifying the effectiveness of ULS.

PLC: To verify the validity of the added PLC, we remove it and only retain the classification head, which is named with w/o (i.e., without) PLC method. The results are shown in the fourth column of Table II. Compared to the complete DSCA-Net, the three metrics of the w/o PLC method have different degrees of degradation on the two datasets. Numerically, the OA of DSCA-Net is 2.61% and 1.41% higher than the w/o PLC. The comparison results show that dual-stream structure can decouple the generation and utilization process of the pseudo-labels, which is useful for the improvement of classification results.

SGLP: In addition, to validate the effectiveness of SGLP, an equal number of unlabeled samples were randomly selected on the same image for the experiment, which is named with w/o (i.e., without) SGLP. The results are shown in the fifth column of Table II. It can be noticed that the three metrics of the w/o SGLP method are reduced compared to the proposed method. Instead, we select the propagated labeled samples as the unlabeled training set in a balanced manner via SGLP.

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TABLE II
COMPARISON OF QUANTITATIVE METRICS OF ABLATION EXPERIMENTS ON PAVIAU AND HOUSTON 2013 DATASETS

Dataset	Metrics	w/o ULS	w/o PLC	w/o SGLP	w/o CADT	DSCA-Net
PaviaU	OA (%)	90.90±3.04	92.07±2.33	93.94±1.76	93.37±2.50	**94.68±1.48**
	AA (%)	92.12±1.72	91.36±2.55	93.12±1.55	93.27±1.13	**93.81±1.46**
	κ (%)	88.17±3.82	89.58±3.03	92.01±2.30	91.30±3.20	**92.97±1.93**
Houston 2013	OA (%)	87.73±2.39	87.63±2.10	88.58±1.71	88.34±1.97	**89.04±1.96**
	AA (%)	89.23±1.80	89.26±1.74	89.93±2.01	89.73±1.54	**90.37±1.48**
	κ (%)	86.73±2.58	86.63±2.26	87.66±1.91	87.40±2.13	**88.16±2.12**

TABLE III
COMPARATIVE CLASSIFICATION RESULTS FOR EACH CLASS OF THE "W/O CADT" AND THE PROPOSED DSCA-NET METHODS

PaviaU			Houston 2013		
No.	w/o CADT	DSCA-Net	No.	w/o CADT	DSCA-Net
1	82.26±7.96	**86.01±7.70**	1	87.64±4.96	**89.08±5.48**
2	96.15±3.78	**98.15±1.42**	2	**88.53±4.63**	87.03±4.22
3	**97.14±2.47**	96.70±2.49	3	96.38±2.27	**97.16±1.53**
4	**88.00±5.26**	86.18±5.18	4	91.17±2.16	**92.82±3.16**
5	**98.94±1.63**	98.41±1.99	5	97.74±5.11	**99.12±1.76**
6	96.60±2.11	**96.89±3.83**	6	88.98±6.27	**89.21±7.69**
7	97.97±2.11	**98.74±1.88**	7	87.78±7.64	**89.56±5.19**
8	**95.02±2.33**	94.23±3.27	8	65.27±6.27	**67.19±10.04**
9	87.33±5.24	**88.94±6.63**	9	84.76±5.62	**85.67±5.03**
			10	90.20±12.0	**91.33±12.82**
			11	87.07±8.56	**87.89±9.18**
			12	**87.12±8.07**	85.05±9.50
			13	96.54±2.62	**96.75±1.73**
			14	99.69±0.93	**100±0.00**
			15	97.12±3.89	**97.71±3.50**

In this way, the negative effects of bias generated during the training process and polarization of the prediction results can be effectively mitigated.

CADT: We fix the threshold $\tau = 0.95$ for all classes, namely w/o (i.e., without) CADT, and compare it with the CADT strategy proposed. The results are shown in the sixth row of Table II. It can be seen that our method outperforms "w/o CADT" in different objective metrics for PaviaU and Houston 2013 datasets. Furthermore, to illustrate the effect of the CADT module in improving classification performance, Table III shows comparative classification results for each class of the "w/o CADT" and the proposed DSCA-Net methods on these datasets. It can be seen that the DSCA-Net has improved the majority of the classes in both datasets, i.e., 5 out of 9 for the PaviaU dataset, and 13 out of 15 for the Houston 2013 dataset. Especially, the discrimination capability in recognizing similar land covers can also be enhanced compared to the "w/o CADT" method. Taking the Houston 2013 dataset as an example, the "Road" and "Highway" classes have high similarity and tend to be misclassified with each other. After the introduction of the CADT module, the classification accuracy of these similar land covers has improved from 84.76% to 85.67% and 90.20% to 91.33%, respectively.

E. Performance Comparison

To evaluate the effectiveness of DSCA-Net, we compare DSCA-Net with other advanced HSI classification methods, including a traditional SVM [46] method, four fully supervised methods based on DL (S3Net [47], RSSAN [48], A2S2K [49], and MTGAN [50]), and two semi-supervised paradigm methods (Semi-CNN [51] and DMSGer [52]). The DSCA-Net and all DL-based methods were executed with PyTorch. Next, the above methods will be briefly introduced.

SVM: This is a traditional approach to realize HSI classification in the original hyperdimensional feature space via support vector machine (SVM). All involved hyperparameters are following the default values.

S3Net: This is a two-branch spectral–spatial Siamese network, where differentiated patches are fed into the model and they can provide more features to the model to improve the recognition ability. The hyperparameter settings followed the optimal values in the article.

RSSAN: This is a fully supervised patchwise-based method that embeds a continuous spectral–spatial attention module in a residual block for adaptive feature refinement of spectral–spatial features learning. For uniformity, the learning rate is set to 1e-4 and the kernel numbers are set to 32 in this method.

A2S2K: This method combines spectral attention with improved 3-D ResBlocks in a fully supervised way and uses an efficient feature recalibration mechanism to improve the classification performance. To unify, the patch size is set to 11×11.

MTGAN: This is a fully supervised paradigm multitask GAN that takes on both reconstruction and classification tasks to improve the discrimination and generalization of classification tasks. For uniformity, the fusion weights are set to 0.33 on all datasets.

Semi-CNN: This method adds skip connection layers between the encoder and decoder to allow the network to be trained to simultaneously minimize the sum of supervised and unsupervised loss functions, thus making the network suitable for SSI. The learning rate is set to 1e-3 and the number of epoch is set to 40.

DMSGer: This method builds a relatively small graph at the region-level based on the superpixel segmentation technique and metric learning, which constitutes SSI to reduce the reliance on labeled samples and captures the pixel-level and region-level information in one graph convolution layer. As to the number of segmented regions, it is set to 50 on the PaviaU dataset and 12 on the other datasets.

Table IV shows OA, AA, κ, and classification accuracies for all categories on the PaviaU dataset. For each line, the bold font indicates the best result. In general, the proposed method DSCA-Net achieves the best results in terms of three objective evaluation metrics. Specifically, as shown in Fig. 5

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TABLE IV
CLASSIFICATION RESULTS FOR PAVIAU DATASET WITH TEN LABELED SAMPLES PER CLASS

No.	SVM	Semi-CNN	DMSGer	RSSAN	A2S2K	S3Net	MTGAN	DSCA-Net
1	65.61±5.56	54.86±8.29	62.58±8.53	70.10±8.82	81.54±5.32	**90.44±7.55**	89.86±4.81	86.01±7.70
2	67.94±7.61	89.49±3.82	80.16±8.25	91.60±4.21	89.04±4.56	83.90±7.09	92.36±5.54	**98.15±1.42**
3	67.54±11.39	78.38±9.90	88.74±6.04	62.81±5.19	88.45±8.95	82.75±8.24	83.83±9.02	**96.70±2.49**
4	85.37±5.93	**95.74±1.28**	80.86±8.21	88.60±6.13	94.74±2.40	94.30±2.40	88.84±7.83	86.18±5.18
5	99.15±0.47	99.75±0.21	**99.95±0.11**	98.80±2.62	99.86±0.11	99.82±0.13	99.07±0.99	98.41±1.99
6	64.81±7.78	57.22±5.37	92.10±7.25	67.05±14.48	73.13±11.49	89.45±8.37	89.74±9.04	**96.89±3.83**
7	89.95±3.90	84.45±4.56	96.30±3.72	83.61±8.85	96.75±3.15	**99.70±0.28**	94.03±5.50	98.74±1.88
8	70.88±11.46	50.69±9.67	83.26±6.57	66.30±9.00	88.04±7.49	92.94±4.47	**94.88±2.47**	94.23±3.27
9	**99.84±0.11**	99.35±0.55	92.20±5.59	98.99±1.04	98.36±4.18	98.16±1.85	92.88±5.75	88.94±6.63
OA (%)	71.05±3.79	77.27±2.04	80.96±4.14	81.72±2.69	87.08±1.87	88.33±2.27	91.48±2.63	**94.68±1.48**
AA (%)	79.01±2.17	78.88±1.28	86.24±2.09	80.87±2.67	89.99±1.91	92.38±1.21	91.72±1.76	**93.81±1.46**
κ (%)	63.48±4.29	70.27±2.56	75.73±4.83	75.80±3.58	83.09±2.38	84.99±2.69	88.83±3.37	**92.97±1.93**

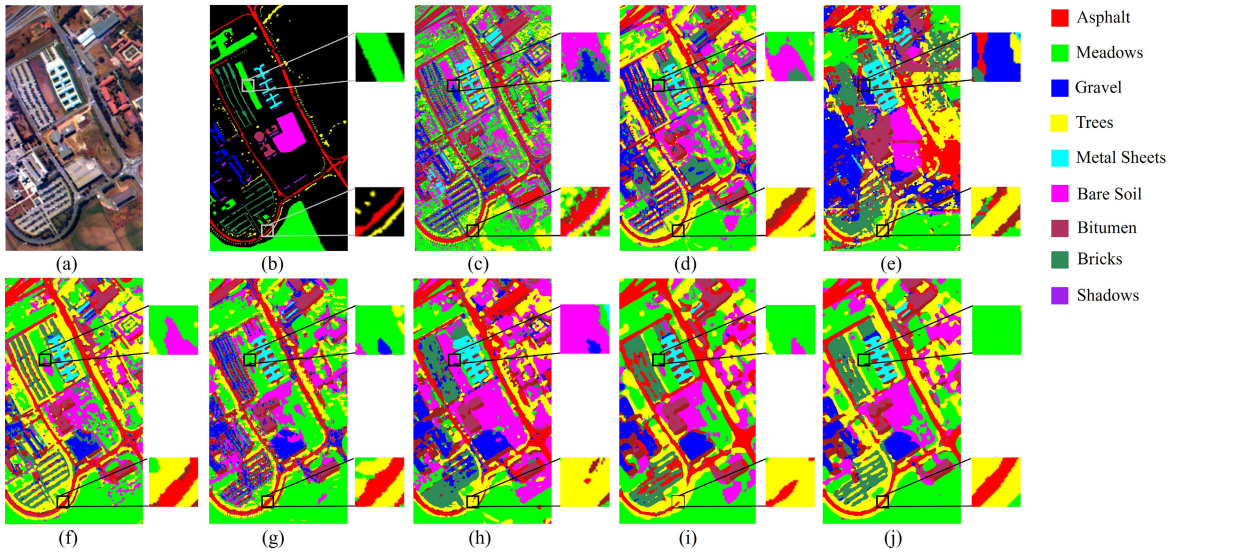


Fig. 5. Classification maps for the PaviaU dataset. (a) False color image. (b) Ground-truth. (c) SVM. (d) Semi-CNN. (e) DMSGer. (f) RSSAN. (g) A2S2K. (h) S3Net. (i) MTGAN. (j) DSCA-Net.

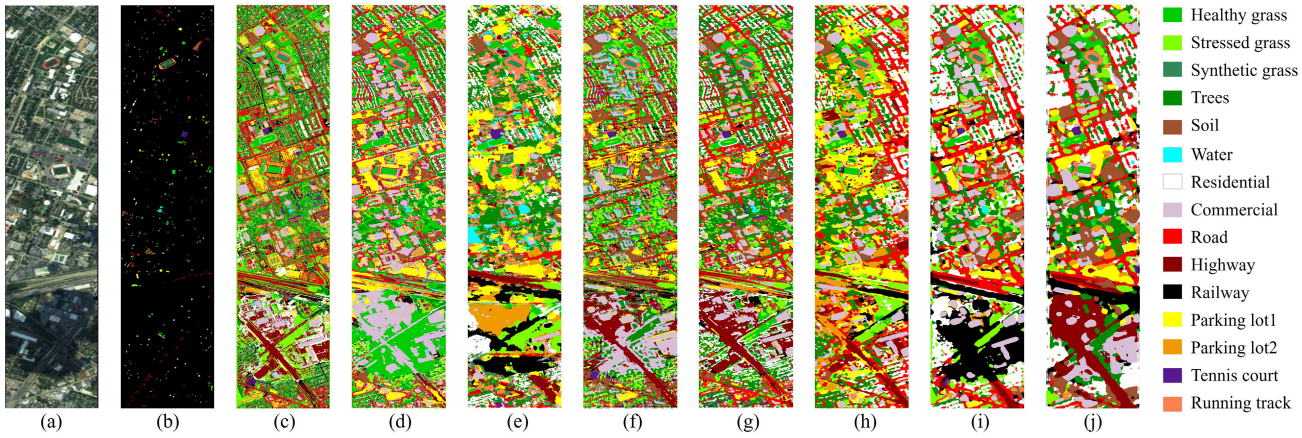


Fig. 6. Classification maps for the Houston 2013 dataset. (a) False color image. (b) Ground-truth. (c) SVM. (d) Semi-CNN. (e) DMSGer. (f) RSSAN. (g) A2S2K. (h) S3Net. (i) MTGAN. (j) DSCA-Net.

in the top enlarged key area, many methods have misclassified a large landform that belongs to the "Meadow" class in the ground-truth into "Gravel," "Bare Soil," or "Trees" class, whereas accurate classification of this area is achieved in the proposed DSCA-Net. Second, as shown in the ground-truth, the "Asphalt" and "Trees" classes should be hierarchical and contiguous in the bottom-zoomed key area. However, many methods confused these two classes, resulting in irregular or even discontinuous asphalt pavements. In contrast, DSCA-Net precisely recognizes the hierarchical and continuous "Asphalt" and "Trees" classes in the bottom-zoomed area. It demonstrates that DSCA-Net enhances the distinguishability among various classes by setting class-adaptive thresholds.

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TABLE V
CLASSIFICATION RESULTS FOR HOUSTON 2013 DATASET WITH TEN LABELED SAMPLES PER CLASS

No.	SVM	Semi-CNN	DMSGer	RSSAN	A2S2K	S3Net	MTGAN	DSCA-Net
1	87.24±6.42	**94.88±6.84**	78.29±9.45	88.72±4.86	93.85±5.92	88.07±6.18	85.09±6.61	89.08±5.48
2	88.97±7.10	52.85±10.94	77.32±13.34	84.05±12.63	91.42±6.78	**96.01±5.41**	88.07±7.97	87.03±4.22
3	98.22±1.28	93.23±3.49	99.17±1.32	96.59±4.81	**99.78±0.51**	99.77±0.23	98.43±1.72	97.16±1.53
4	92.55±3.34	88.46±6.51	71.08±12.11	85.46±6.93	**94.15±4.06**	93.03±9.02	93.83±2.29	92.82±3.16
5	89.86±6.57	**99.54±0.47**	98.56±2.02	91.70±5.88	98.62±1.32	94.42±8.09	98.56±1.34	99.12±1.76
6	83.46±1.64	88.41±3.07	**94.51±6.25**	78.57±6.95	92.00±5.56	92.98±5.27	86.10±6.24	89.21±7.69
7	73.42±11.17	77.03±11.29	69.87±16.02	71.03±7.48	81.94±13.05	82.22±9.66	**90.12±6.85**	89.56±5.19
8	43.63±7.81	52.63±17.49	46.37±5.64	45.26±7.77	56.46±12.43	56.52±6.04	53.36±10.70	**67.19±10.04**
9	77.22±5.13	65.02±4.05	68.83±7.22	71.80±5.74	78.87±7.99	74.63±10.30	**86.10±6.24**	85.67±5.03
10	41.41±10.32	56.79±16.89	78.81±12.76	72.79±9.85	77.26±13.86	76.29±12.88	72.91±20.60	**91.33±12.82**
11	64.48±8.01	36.80±15.97	78.94±7.07	54.41±7.29	74.53±9.31	86.59±10.70	76.06±10.17	**87.89±9.18**
12	33.15±9.41	67.09±17.17	71.17±12.55	79.79±5.99	73.22±14.13	61.46±14.27	83.25±16.86	**85.05±9.50**
13	28.85±10.66	94.60±2.65	93.55±5.46	60.39±15.21	93.99±1.60	92.83±2.41	92.79±2.86	**96.75±1.73**
14	95.57±3.71	88.61±6.85	**100±0.00**	90.57±4.19	99.69±0.55	**100±0.00**	98.54±1.94	**100±0.00**
15	98.88±0.25	99.03±1.34	99.65±0.54	92.71±5.26	99.06±1.38	**99.86±0.42**	98.74±1.23	97.71±3.50
OA (%)	71.72±1.68	73.32±2.30	77.97±2.53	76.43±2.12	84.70±1.81	83.81±1.62	85.06±2.94	**89.04±1.96**
AA (%)	73.13±1.34	77.00±2.00	81.74±1.96	77.59±2.31	86.99±1.39	86.31±1.34	86.83±2.41	**90.37±1.48**
κ (%)	69.44±1.82	71.19±2.49	76.21±2.73	74.53±2.29	83.47±1.95	82.52±1.75	83.85±3.18	**88.16±2.12**

TABLE VI
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON THE OTHER TWO DATASETS

Dataset	Metrics	SVM	Semi-CNN	DMSGer	RSSAN	A2S2K	S3Net	MTGAN	DSCA-Net
KSC	OA (%)	72.76±6.29	79.02±1.73	95.41±2.43	86.74±2.59	89.76±1.61	94.58±2.58	96.05±1.35	**99.59±0.23**
	AA (%)	66.30±7.57	73.89±1.96	95.02±2.57	81.17±3.30	84.02±2.65	92.66±2.96	92.89±2.28	**99.19±0.50**
	κ (%)	69.66±7.01	76.63±1.92	94.89±2.70	85.19±2.89	88.59±1.79	93.97±2.86	95.59±1.51	**99.54±0.25**
Botswana	OA (%)	87.04±0.93	96.29±1.02	98.57±0.90	96.42±1.27	98.75±0.61	95.41±0.94	96.89±1.44	**99.08±0.70**
	AA (%)	88.78±0.80	96.56±0.94	98.73±0.79	96.33±1.49	98.81±0.65	98.55±0.80	97.09±1.35	**99.03±0.79**
	κ (%)	85.97±1.00	95.98±1.10	98.45±0.98	96.12±1.38	98.65±0.66	98.28±1.02	96.63±1.56	**99.00±0.76**

Table V shows the objective metrics for each method for the Houston 2013 dataset. For each line, bold text indicates the best performance. Numerically, the proposed method obtains the highest OA value (89.04%), while the Semi-CNN obtains the lowest OA value (73.32%). DSCA-Net can obtain the highest classification accuracy values in 6 of the total 16 categories, which also proves the effectiveness of the proposed method. The classification maps for the Houston 2013 dataset are shown in Fig. 6. Although the compared methods can obtain good classification results by combining the spatial–spectral features of HSI, misclassification still occurs due to the small number of labeled samples, resulting in point-like noise on the classification maps. In contrast, the proposed DSCA-Net method can expand the training set by pseudo-labeling the unlabeled samples in a class-adaptive manner, which improves the generalization ability and discrimination of the model.

In addition to the previous PaviaU and Houston 2013 datasets, we also evaluate the classification performance of different methods on two other well-known datasets, i.e., KSC and Botswana. Here, we also show the OA, AA, and κ values of the proposed and compared methods for the two different datasets. The results with ten labeled samples per class are shown in Table VI. For each line, the bold font indicates the best result. It is worth noting that the A2S2K method achieves the second-best result on the Botswana dataset, but not significantly on the remaining dataset. A similar situation also exists for the RSSAN method. In contrast, our proposed method DSCA-Net always outperforms the other comparative methods on both two datasets. It indicates that the generalization and robustness of our proposed method reach a high level.

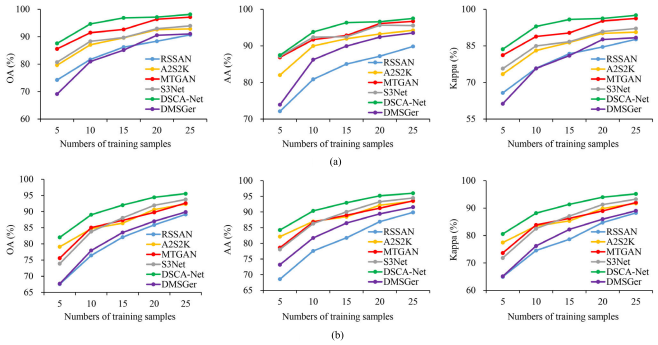


Fig. 7. Overall accuracies with respect to different numbers of training samples on two datasets. (a) PaviaU. (b) Houston 2013.

F. Effect of Training Sample Numbers

Furthermore, we evaluate the classification performance with different numbers of training samples. Fig. 7 shows the overall classification accuracy of the proposed method and the comparison method. In each category, the number of training samples varies from 5 to 25, increasing gradually with a step of 5. In general, the results of all methods trend higher as more labeled samples are used for training. This is reasonable due to more samples can help to learn more discriminative DL models for higher prediction accuracy. Besides, we can also notice that continuously higher classification results can be obtained by the proposed DSCA-Net, in terms of OA, AA, and

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κ metrics. Furthermore, the advantage is more obvious when a small training sample set is used. Overall, the experimental results show that the proposed method has better robustness and generalization ability, especially in the case of limited samples.

IV. Conclusion

We propose a novel dual-stream class-adaptive semi-supervised network (DSCA-Net) in this article. By designing a dual-stream structure with the consistency regularization concept, limited labeled samples and massive unlabeled samples cooperate to learn discriminative feature representation, thus improving the classification performance. To solve two issues within the dual-stream semi-supervised network, i.e., imbalanced data distribution and diverse classification difficulties for different classes, we design an SGLP module to approximately construct a category-balanced unlabeled set, as well as developing a CADT strategy to assign various thresholds for different classes. In this manner, reliable unlabeled samples are adaptively injected into the model training. Both theoretical analysis and experimental results have demonstrated the effectiveness of the DSCA-Net over some state-of-the-art methods, especially for very few training samples.

Although the proposed DSCA-Net introduces unlabeled sample information to improve the semi-supervised classification results, it is limited to single-scene data and ignores the domain shifts in cross-scene classification tasks. In future work, we can consider better criteria to divide the "hard" and "easy" classes and introduce cross-domain learning in the proposed framework to make the network applicable to different HSIs.

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